

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$(a) \begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_1: \sigma_M^2 \neq \sigma_F^2 \end{cases}$$

$$N = 1000 \quad df_M = 577 - 4 = 573 \quad df_F = (1000 - 577) - 4 = 419$$

$$\begin{cases} \hat{\sigma}_M^2 = \frac{SSE_M}{df_M} \approx 169.561 \\ \hat{\sigma}_F^2 = (12.024)^2 \approx 144.571 \end{cases}$$

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.561}{144.571} \approx 1.1708$$

$$F_{0.05, 573, 419} \approx 1.168$$

$$\Rightarrow F > 1.168 \Rightarrow \text{reject } H_0$$

- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

$$(b) \begin{cases} H_0: \sigma_{SINGLE}^2 = \sigma_{MARRIED}^2 \\ H_1: \sigma_{SINGLE}^2 < \sigma_{MARRIED}^2 \end{cases}$$

$$df_{SINGLE} = 400 - 5 = 395, \quad df_{MARRIED} = 600 - 5 = 595$$

$$\begin{cases} \hat{\sigma}_{SINGLE}^2 = (56231.0382) / 395 \approx 142.3571 \\ \hat{\sigma}_{MARRIED}^2 = (100703.0471) / 595 \approx 169.2488 \end{cases}$$

$$F = \frac{\hat{\sigma}_M}{\hat{\sigma}_S} = \frac{169.2488}{142.3571} \approx 1.1889 \quad F_{0.05, 395, 595} \approx 1.163$$

$$F > 1.163 \Rightarrow \text{reject } H_0$$

- c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

$$NR^2 = 59.03 \quad \chi^2_{(0.05, 4)} \approx 9.488$$

$$\therefore NR^2 > \chi^2_{(0.05, 4)} \Rightarrow \text{reject } H_0$$

No, the married variable was not included in this regression

- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$\therefore \begin{cases} \text{original variable} : EDUC, EXPER, METRO, FEMALE \\ \text{squared variable} : EDUC^2, EXPER^2 \\ \text{cross-products} : EDUC \times EXPER, EDUC \times METRO, EDUC \times FEMALE, \\ \quad EXPER \times METRO, EXPER \times FEMALE, METRO \times FEMALE \end{cases}$$

$$\therefore df = 12 \Rightarrow \chi^2_{(0.05, 12)} = 21.026 \quad \therefore 78.82 > 21.026$$

$\therefore$ reject H_0 , heteroskedasticity exists

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

narrower : EXPER, METRO, FEMALE

wider : EDUC

No, there is no inconsistency. The mathematical formula for white's rbse allows them to be either larger or smaller than usual OLS s.e. in any given finite sample.

- f. If we add *MARRIED* to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

The results are completely compatible

$\therefore$ (f) shows that marital status doesn't shift the average level of person's wage

(b) shows that marital status change the spread or dispersion of a person's wage around the average

$\therefore$ it is entirely possible